

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths *	Prac **
3	(a)	(i)		Any four from: A. Description of direction of fluid /countercurrent flow – maintains concentration gradient (1) B. replacement of dialysis fluid – maintains concentration gradient (1) C. Temperature/ 40 °C - increase rate of diffusion/maintains patient temperature/ prevents temperature shock (1) D. large number of tubes - increase surface area (1) E. pore size/ selectively permeable – prevents loss of (plasma) proteins(1)		2	2	4		
	(b)	(i)		27		1		1		
		(ii)		Posterior pituitary (gland)	1			1		
		(iii)		Any five from A. ADH makes membranes of {collecting duct/distal convoluted tubule} more permeable to water (1) B. ADH combines with receptor (1) C. Vesicles fuse with membrane on <u>collecting duct</u> side (1) D. Inserting <u>more</u> aquaporin into membrane (1) E. Water potential of filtrate is higher than water potential within {cell/ tissue fluid/ blood} OWTTE (1) F. Water moves into {epithelial cell / tissue fluid} by osmosis/ water leaves filtrate} by osmosis (1)		2	3	5		
	(c)			<u>Membrane</u> not made more permeable so more water stays in {filtrate/urine} / no aquaporins inserted into <u>membrane</u> so less water reabsorbed		1		1		
	(d)			Nn x nn (1) correct offspring genotypes (1) 1 in 2 / 0.5 / 50% / ½ (1) If incorrect parental genotypes can award second mark only for correct working Reject 1:1		3		3		
				Question 3 total	1	9	5	15	0	